# Certified Robustness from One Setpoint Step: A Safe Acceptance Test for Model-Free PID Tuning

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***Abstract*— Model-free PID tuners improve a loop from closed-loop data, but none guarantees its robustness, measured by the peak $M_s$ of the sensitivity function. We show that the setpoint step these tuners already apply is enough to certify it. In a stable digital loop, the increments of the error after a unit setpoint step are the impulse response of the sensitivity function. One finite record therefore gives a lower and an upper bound on $M_s$, under one assumption on the unrecorded rest of the response. The same record also certifies the stability and the $M_s$ of any other controller before it is applied. With these two certificates, any iterative tuner can accept only tunings with $M_s \le M^*$, and never apply a tuning with $M_s > M^*$. We add them to SPIN, a tuner that counts the turns of the error in phase portraits, in two steps: certified SPIN checks each tuning after applying it, safe SPIN before. On $19$ simulated loops with $M^* = 1.7$, basic SPIN ended above $M^*$ on up to $9$ loops and applied tunings above $M^*$ in $57\%$ of its iterations; certified SPIN ended below $M^*$ on all loops, but still applied such tunings in $45\%$; safe SPIN never applied one.**

## I. Introduction

Let $C$ be the controller, $P$ the plant, $L = CP$ the loop transfer function, $S = 1/(1+L)$ the sensitivity function and $T = 1 - S$ the complementary sensitivity function. The peak $M_s = \max_\omega |S|$ measures robustness: $1/M_s$ is the shortest distance from the Nyquist curve of $L$ to the critical point $-1$, so $M_s$ near $1.4$ is safe and near $2$ is poor [1]. Model-based tuning rules reach a chosen $M_s$ [1], [2]. Model-free tuners adjust the gains from closed-loop data instead, by iterative feedback tuning [3], extremum seeking [4], virtual reference feedback tuning [5], setpoint overshoot [6] or step-response inspection [7]. They target performance or a response shape; none guarantees the $M_s$ of its result, or of the tunings it tries on the way. Once a loop runs, its $M_s$ is rarely checked at all [8], [9].

The $\mathcal{H}_\infty$ norm of a system, and hence $M_s$, can be estimated from data: by power iterations over repeated dedicated experiments [10], [11], by data-driven estimation [12], from limit cycles induced in closed loop [13], or from impulse-response data under a decay assumption, with a one-sided guarantee [14]; margins can be bounded from data [15]. These methods need dedicated experiments, give estimates or one-sided bounds, and are not built into a tuner. Safe Bayesian optimization tunes controllers without leaving a safe set [16], with probabilistic guarantees under a Gaussian-process model of the cost.

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TABLE I
Main symbols

| | |
|---|---|
| $C$, $P$, $L = CP$ | controller, plant, loop transfer function |
| $S = \frac{1}{1+L}$, $T = 1 - S$ | sensitivity, complementary sensitivity |
| $M_s = \max \lvert S \rvert$, $M^*$ | robustness measure, its target |
| $e_n$, $a_n = e_n - e_{n-1}$ | error after a unit setpoint step, its increments |
| $\hat{S}(\theta)$ | estimate of $S$ from the record $e_0, \dots, e_N$ |
| $\tau$ | bound on the unrecorded tail $\sum_{n>N} \lvert a_n \rvert$ |
| $\overline{M}$ | certified upper bound on $M_s$, as computed |
| $\rho = C'/C - 1$ | relative change from $C$ to the next controller $C'$ |
| $\Gamma_k$, $N_k$, $\bar{N}_k$ | SPIN portraits, their turn counts, limits |

*The idea.* A tuner already applies a setpoint step in every iteration. After a unit step, the increments $a_n = e_n - e_{n-1}$ of the error $e$ are the impulse response of $S$ (Lemma 1), so their Fourier sum is $S$ itself. One record thus shows $|S|$ at every frequency, and with it the peak $M_s$ (Fig. 1). We turn this elementary fact into guarantees, for a chosen target $M^*$:

1) *bounds*: one finite record gives a lower and an upper bound on $M_s$ that hold with certainty, not estimates (Section II, Theorem 1);
2) *tests for any iterative tuner*: a test that passes only tunings with $M_s \le M^*$ (Theorem 2), and a test of the next tuning *before* it is applied (Theorem 3), which keeps every applied tuning below $M^*$ (Section III);
3) *application to SPIN* [7]: two variants built on the tests, certified SPIN, which ends below $M^*$, and safe SPIN, which never applies a tuning above $M^*$ (Section IV, Corollaries 1 and 2), and why its turn counts alone cannot certify $M_s$ (Lemma 2, Fig. 2).

Section V confirms the results in simulation. Table I lists the main symbols.

## II. A Certificate from One Setpoint Step

A digital controller runs with the sampling period $T_s$. Seen at the sampling instants, the loop it closes is linear and time invariant, with transfer functions $C(z)$, $P(z)$, $L$, $S$ and $T$ as above. On the unit circle, $z = \mathrm{e}^{j\theta}$ with the normalized frequency $\theta = \omega T_s \in [0, \pi]$, and $M_s = \max_\theta |S(\mathrm{e}^{j\theta})|$; we write $M_s(C)$ when the controller matters. All these quantities refer to the digital loop, the loop that actually runs; e (upright) is Euler's number.

*Lemma 1 (The step contains $S$):* Let the loop be stable and have one degree of freedom (the controller acts on the error $e = r - y$, with setpoint $r$ and output $y$), and let $L$ be strictly proper (the measurement precedes the control action). Let $e_n$ be the error after a unit setpoint step at $n = 0$, applied from rest and without disturbances, so that $e_{-1} = 0$ and

$e_0 = 1$, and let $a_n = e_n - e_{n-1}$. Then

$$S(\mathrm{e}^{j\theta}) = \sum_{n\geq 0} a_n \, \mathrm{e}^{-jn\theta}. \tag{1}$$

*Proof:* Let $R(z)$ and $E(z)$ be the $z$-transforms of the setpoint and the error. A unit step has $R = 1/(1-z^{-1})$, and $E = SR$, so $S = (1 - z^{-1})E$, whose coefficients are the increments $a_n$. In a stable loop, $a_n$ decays exponentially, so the series converges on the unit circle. □

A real record $e_0, \ldots, e_N$ is finite, so one assumption covers the increments it does not show.

*Assumption 1 (Long enough record):* The loop is stable, and the record $e_0, \ldots, e_N$ satisfies $\sum_{n>N} |a_n| \leq \tau$ for a known $\tau$.

This is the only assumption. A valid $\tau$ shrinks geometrically as the record gets longer (Section V checks it with a four times longer record). No finite record can prove that the loop is stable: an unstable or slow mode may appear only after the record ends. So the certificate bounds $M_s$ of a loop assumed stable, and Theorem 3 later certifies the stability of each next controller from the record of the current one.

The record gives the estimate $\hat{S}(\theta) = \sum_{n=0}^{N} a_n \mathrm{e}^{-jn\theta}$ of $S$.

*Theorem 1 (Certified bounds from one record):* Under Lemma 1 and Assumption 1, $|S(\mathrm{e}^{j\theta}) - \hat{S}(\theta)| \leq \tau$ at every frequency, so

$$\max_\theta |\hat{S}(\theta)| - \tau \;\leq\; M_s \;\leq\; \max_\theta |\hat{S}(\theta)| + \tau. \tag{2}$$

*Proof:* By (1), $S - \hat{S} = \sum_{n>N} a_n \mathrm{e}^{-jn\theta}$, whose magnitude is at most $\sum_{n>N} |a_n| \leq \tau$. □

*Computing the bound.* A computer evaluates $\hat{S}$ on a finite grid of frequencies, all at once by one chirp $z$-transform, an FFT-type algorithm. Two simple margins keep the result certified. Between grid points, $|\hat{S}|$ can rise only by the grid spacing times a slope bound, a weighted sum of the $|a_n|$. At high frequencies, where the grid would have to be very fine, summation by parts bounds $|S - 1|$ by the total variation of the record divided by $\sin(\theta/2)$. We call $\overline{M}$ the largest grid value of $|\hat{S}|$ plus $\tau$ plus these margins; like (2), $M_s \leq \overline{M}$ holds with certainty. Fig. 1 shows the bounds on the example that runs through the paper: plant $P_3$ of Table II with PID, tuned by basic SPIN.

## III. A Certified Acceptance Test

An iterative tuner applies a controller $C$, records the error after a setpoint step, and proposes the next controller from the record. We call a set of gains a *tuning*. The *test* adds one check to any tuner: a tuning *passes* if the upper bound $\overline{M}$ of Theorem 1, computed from its own record, is at most the target $M^*$. A tuning that fails is lowered, by the tuner's own rule or by a *uniform back-off* $C \to (1 - \beta)C$, $0 < \beta < 1$, which scales all gains down by the same factor. A tuner *accepts* a tuning when it keeps it as a candidate result; with the test, it accepts only tunings that pass.

From here on, the plant is stable (so its static gain $P(1)$ is finite), as are the self-regulating processes that SPIN targets, and the controller has integral action, $C = C_0 + K_i T_s/(z-1)$

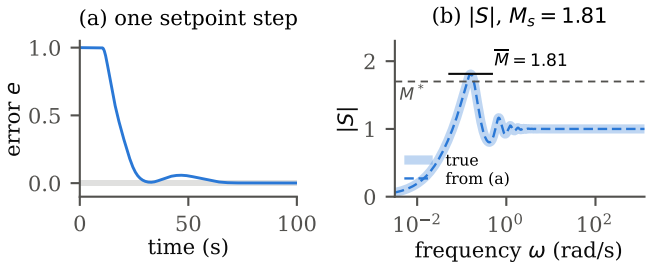



Fig. 1. One step certifies $M_s$, on the running example: $P_3$ with PID under the final tuning of basic SPIN. (a) The error after a unit setpoint step; it never crosses the setpoint (grey band: ±2%). (b) $|\hat{S}|$ from the record lies on the true $|S|$; the certified bounds, $1.809 \leq M_s \leq \overline{M} = 1.813$ ($\tau = 10^{-3}$), show $M_s > M^* = 1.7$: the tuning fails the test.

with $K_i > 0$ and a stable proportional and derivative part $C_0$. We choose the margins of $\overline{M}$ so small that a tuning with $M_s \leq M^* - 2\tau - \eta$ always passes, for a small tolerance $\eta$.

*Theorem 2 (Certified acceptance):* Let every record meet Lemma 1 and Assumption 1. Then every tuning that passes the test has $M_s \leq M^*$, whatever rule proposed it.

*Proof:* Passing means $\overline{M} \leq M^*$, and $M_s \leq \overline{M}$. □

So the test never passes a fragile tuning. Nor can it trap a tuner: a uniform back-off shrinks the Nyquist curve of $L$ towards the origin, away from $-1$, so it always leads to a tuning that passes.

*Proposition 1 (A back-off restores the test):* Let $K_i P(1) > 0$ (the integral action counteracts the error), and let $\mu \leq 0$ be the leftmost real part of the Nyquist curve of $C$ ($\mu = 0$ if it has none below zero). For $0 < \lambda < 1/|\mu|$, the controller $\lambda C$ stabilizes the plant, and

$$M_s(\lambda C) \leq \frac{1}{1 - \lambda|\mu|}. \tag{3}$$

So, if $M^* > 1 + 2\tau + \eta$, finitely many uniform back-offs lead to a tuning that passes.

*Proof:* At every frequency, $|1 + \lambda L| \geq 1 + \lambda \operatorname{Re} L \geq 1 - \lambda|\mu| > 0$, which gives (3). ($\mu$ is finite: near $z = 1$, the integral term has the real part $-K_i T_s/2$.) For stability, apply the Nyquist criterion on the unit circle with a small indentation outside the pole $z = 1$, where $\lambda L$ has a positive real part since $K_i P(1) > 0$: $\lambda L$ has no poles outside this contour, and its curve stays in $\operatorname{Re} \geq -\lambda|\mu| > -1$, so it cannot encircle $-1$. Back-offs drive $\lambda$ to zero, and with it the bound (3) towards 1, until $M_s \leq M^* - 2\tau - \eta$ and the tuning passes. □

The test checks a tuning only after it has been applied: the tuner learns that a tuning is fragile only by running it. But the record of $C$ gives $S$ of the running loop, and $S$ determines the sensitivity of any other controller $C'$ on the same plant. Let $\rho = C'/C - 1$ be the relative change of the controller, so that $L' = L(1 + \rho)$. Then $1 + L' = (1 + L)(1 + T\rho)$, and

$$S' = \frac{S}{1 + T\rho}. \tag{4}$$

This is exact: it predicts $S'$ from $S$ and the planned change $\rho$, without a model of the plant. The record gives $\hat{S}$ and $\hat{T} = 1 - \hat{S}$, each within $\tau$.

*Theorem 3 (Certify before applying):* Let $C$ and $C'$ have integral action and the same $T_s$, and let the record of $C$ meet Lemma 1 and Assumption 1. If, at every frequency and for every $0 \le \lambda \le 1$,

$$|1 + \lambda \hat{T}\rho| > \tau|\rho|, \tag{5}$$

then $C'$ stabilizes the plant, and

$$M_s(C') \le \max_{\theta} \frac{|\hat{S}| + \tau}{|1 + \hat{T}\rho| - \tau|\rho|}. \tag{6}$$

In words: (5) says that, as the controller moves gradually from $C$ to $C'$, the Nyquist curve never passes through $-1$; the bound (6) is (4) with the record error $\tau$ added.

*Proof:* Move from $C$ to $C'$ through $C_\lambda = C(1 + \lambda\rho)$, $0 \le \lambda \le 1$; each has integral action and a stable $C_0$ part. By (4), $1 + L_\lambda = (1 + L)(1 + \lambda T\rho)$, and since $|T - \hat{T}| = |S - \hat{S}| \le \tau$, (5) gives $|1 + \lambda T\rho| > 0$. So $1 + L_\lambda$ never vanishes on the indented unit circle of Proposition 1, the number of its encirclements cannot change with $\lambda$, and the loop, stable at $\lambda = 0$, is stable at $\lambda = 1$. The bound (6) follows from (4), $|S| \le |\hat{S}| + \tau$ and $|1 + T\rho| \ge |1 + \hat{T}\rho| - \tau|\rho|$. □

Like $\overline{M}$, (5) and (6) are evaluated on the grid with margins. Since (4) is exact, Theorem 3 needs no small-step assumption. With $\tau = 0$, (6) becomes $\max_\theta |\hat{S}/(1 + \hat{T}\rho)|$, the *prediction* of $M_s(C')$.

## IV. Application to SPIN

### A. The SPIN method

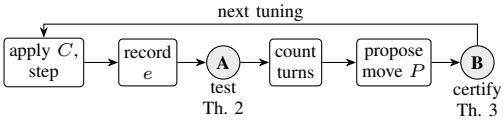
SPIN [7] tunes an I, PI or PID controller with the integral, proportional and derivative gains $K_i$, $K_p$, $K_d$. In each iteration, it applies one setpoint step, records the error $e_n$, and forms three phase portraits, curves in the plane traced as $n$ grows (the Pachner plots of [7]):

$$\begin{aligned} &\Gamma_0 : \big(e_n, \textstyle\sum_{m \le n} e_m\big), \qquad \Gamma_1 : (\Delta e_n, e_n), \\ &\Gamma_2 : (\Delta^2 e_n, \Delta e_n), \end{aligned} \tag{7}$$

with the differences $\Delta e_n = e_n - e_{n-1} = a_n$ and $\Delta^2 e_n = \Delta e_n - \Delta e_{n-1}$. $\Gamma_0$, $\Gamma_1$ and $\Gamma_2$ emphasize low, middle and high frequencies, where the integral, proportional and derivative actions dominate the controller; we call them bands 0, 1 and 2. Each axis is divided by its peak, and the count $N_k$ is the number of turns that $\Gamma_k$ makes around its end point until the error stays within $\pm\delta$ of the setpoint ($\delta$ is 2% of the step). In short, SPIN raises all gains while every portrait turns little, and otherwise lowers the gain of the lowest band that turns too much; Algorithm 1 states the rule.[1]

### B. From certified to safe SPIN

Algorithm 1 adds the tests of Section III in two steps. *Certified SPIN accepts a tuning only if it passes both the counts and the test.* It adds layer A: the counts shape the tuning, and the test decides whether it is accepted. A tuning that passes the counts but fails the test is lowered by the band move that (4) predicts to lower $M_s$, or else by a back-off (A2).




**Algorithm 1:** one iteration (top: where the tests A and B sit): from the tuning $C$ in use to the next tuning; **return** ends the iteration. *Basic SPIN* [7]: lines 1–7. *Certified SPIN*: adds layer A, the test (Theorem 2). *Safe SPIN*: adds A1 and layer B, which certifies each move before applying it (Theorem 3).

1: apply a setpoint step with $C$; record the error $e$
2: **if** $e$ does not decay **then return** retreat($C$)
**A1:** $C$ passes the test iff $\overline{M} \le M^*$ (Theorem 1)
**B1:** **if** $C$ fails **then return** the last tuning that passed (back-off($C$) if none)
3: count the turns $N_0$, $N_1$, $N_2$
4: **if** $N_k \le \bar{N}_k$ for every band $k$ **then**
**A2:** **if** $C$ fails **then return** a band move that (4) predicts to lower $M_s$ (back-off($C$) if none)
5: $P \leftarrow$ raise($C$)
6: **else** $P \leftarrow$ band move $k$, the lowest band with $N_k > \bar{N}_k$
7: **return** $P$
**B2:** (replaces 7) **return** the first of $P$, band moves $2, 1, 0$, back-off($C$) that Theorem 3 certifies ($M_s \le M^*$); $C$ if none

*Actions:* each multiplies the gains as listed, with $g = 1/(1 - \beta)$ (·: unchanged). Bands 0, 1, 2 belong to $K_i$, $K_p$, $K_d$; PI has bands 0, 1.

| action | $K_i$ | $K_p$ | $K_d$ | used in |
|---|---|---|---|---|
| raise | $g$ | $g$ | $g$ | 5 |
| band move 0 | $1/g$ | · | · | 6, A2, B2 |
| band move 1 | $g$ | $1/g$ | · | 6, A2, B2 |
| band move 2 | $g$ | $g$ | $1/g$ | 6, A2, B2 |
| back-off | $1/g$ | $1/g$ | $1/g$ | B1, A2, B2 |
| retreat | $1/2$ | $1/4$ | $1/8$ | 2 |

Gains stay within $[0.001, 10]$ times their initial values (the gain box). *e does not decay* if $\max|e_n|$ over the last fifth of the record exceeds half of $\max|e_n|$. *Details:* A2 takes the highest such band, and backs off after 10 band moves without a pass; a tuning that failed B1 is not proposed again. *Accepted* tunings pass line 4 (basic), lines 4 and A1 (certified), or A1 (safe); Section V reports the most aggressive one accepted in the last 60 iterations. Defaults: $\bar{N}_k = (0.5, 0.75, 1.0)$, $\delta = 0.02$, $\beta = 0.1$, $M^* = 1.7$, $\tau = 10^{-3}$, $\eta = 2 \cdot 10^{-3}$.

*Corollary 1 (Certified SPIN):* Under the conditions of Proposition 1, the result of certified SPIN has $M_s \le M^*$, whatever the limits $\bar{N}_k$. If no gain reaches the gain box, the test never traps SPIN: every failed tuning is followed by a tuning that passes after finitely many moves.

*Proof:* The result is an accepted tuning, so it passed the test, and Theorem 2 applies. After at most 10 band moves, A2 backs off, which leads to a tuning that passes after finitely many back-offs by Proposition 1, unless line 2 or 6 moves the gains first. □

Certified SPIN ends safe, but it learns that a tuning is fragile only by applying it. *Safe SPIN certifies every move before applying it.* It adds layer B: it applies the test to every record and returns to the last tuning that passed when one

[1] An online simulator: `https://robopid-simulator.uceeb.cvut.cz`

fails (B1; A2 then never acts), so moves start only from tunings that passed; and it applies a move only if Theorem 3 certifies it (B2).

*Corollary 2 (Safe SPIN):* Let the plant be stable with $K_i P(1) > 0$, let the first tuning that passes the test stabilize it, and let the record of every stable loop meet Assumption 1. Then, from that tuning on, every tuning that safe SPIN applies stabilizes the plant and has $M_s \le M^*$, whatever the limits $\bar{N}_k$.

*Proof:* By induction over the iterations. Let the tuning $C$ in use be stable; by B1, either $C$ passes, so $M_s(C) \le M^*$ (Theorem 2), or the next tuning is the last one that passed, stable with $M_s \le M^*$. If $C$ passes, B2 applies either $C$ itself or a move that Theorem 3, from the record of $C$, certifies to be stable with $M_s \le M^*$. □

### C. Why the turn counts alone cannot certify

The counts see setpoint crossings, and crossings reveal $M_s$ only on special loops. To show this, write a portrait (7) as $(p, q)$, with $q$ the running sum of $p$, shifted so that the end point is the origin. The count is the net counterclockwise angle swept around the origin, divided by $2\pi$. We state the results in continuous time, where $\dot{q} = p$. For $\Gamma_0$, $p = e$, and its zeros are the setpoint crossings.

*Lemma 2 (Half turns):* Let $\dot{q} = p$ with $p$ continuous, let $p(t_1) = p(t_2) = 0$ with $q(t_1)q(t_2) < 0$, and let $p \neq 0$ on $(t_1, t_2)$. Then, on $[t_1, t_2]$, the curve $(p, q)$ turns counterclockwise around the origin by exactly half a turn, and so does its image under any linear map with a positive determinant, such as the per-axis scaling of SPIN.

*Proof:* On $(t_1, t_2)$, $p$ keeps one sign, so the curve stays in one half-plane bounded by the $q$-axis, and $q$ moves monotonically from one ray of the $q$-axis to the opposite ray. The curve avoids the origin, since $p = 0$ only at $t_1$ and $t_2$, where $q \neq 0$. The angle thus changes by exactly $\pi$, counterclockwise, since $q$ increases where $p > 0$ and decreases where $p < 0$. A linear map with a positive determinant maps half-planes to half-planes and opposite rays to opposite rays, and keeps the orientation. □

So each setpoint crossing adds half a turn to $N_0$, and a count sees the crossings that rise above $\delta$. On some loops, crossings do reveal $M_s$. Lambda, IMC and SIMC rules cancel the plant lag by the controller zero [2], which leaves the loop $L(s) = \kappa\, \mathrm{e}^{-sL_d}/(sL_d)$ with the loop gain $\kappa$. Its $M_s$ grows with $\kappa$, and its error never crosses the setpoint while $\kappa \le 1/\mathrm{e}$, that is, $M_s \le 1.39$ (a classical result on this delay equation [17]); beyond, each new crossing above $\delta$ appears at a fixed $M_s$. So the counts rise in steps (Fig. 2(a)), and the limits $\bar{N}_k = (1/2, 3/4, 1)$ of [7] allow 0, 1 and 2 crossings, that is, $M_s \le 1.53$, 1.84 and 2.21. $\Gamma_0$ binds first, so on these loops SPIN stops at $M_s \le 1.53$.

On general loops, this guarantee is lost. The counts leave the steps (in the 3800 count decisions of certified SPIN in Section V, $N_0$ and $N_2$ lay off them in 23 and 69%), and crossings do not bound $M_s$: the basic tuning of Fig. 1 never crosses the setpoint, yet has $M_s = 1.81$ (a PI loop on a first-order plant reaches $M_s = 2.62$ without a crossing above

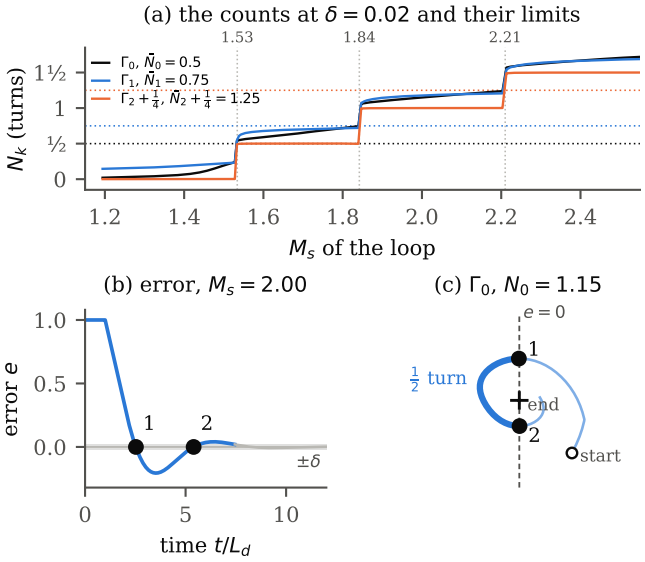



Fig. 2. SPIN's counts on the lag-cancelled loop. (a) Each count steps over its limit (dotted) at $M_s = 1.53$, 1.84, 2.21 ($N_2$ and its limit shifted up a quarter turn). (b) A record with $M_s = 2.00$ and two crossings. (c) Its portrait $\Gamma_0$ turns by exactly half a turn between them (Lemma 2); the radius is compressed for display.

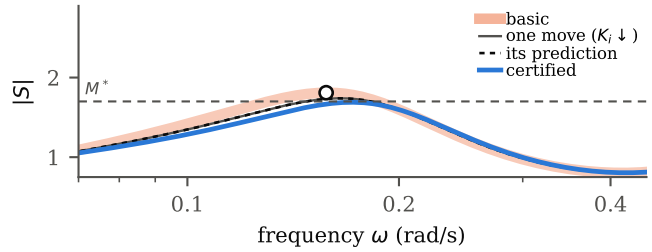



Fig. 3. $|S|$ near its peak, $P_3$ with PID: the basic tuning ($\overline{M} = 1.81$, ○), one band move lowering $K_i$ ($M_s = 1.740$, as predicted by (4) from the record of the basic tuning), and the certified tuning ($M_s = 1.690$).

2%), while a controller zero can cause crossings at a low $M_s$. The counts can shape a tuning, but only a certificate can certify it.

## V. Simulations

*Setup:* We test the three variants on the 19 loops of Table II. Each runs 200 iterations of Algorithm 1 with its defaults. The gain box ties the result to the start, so every loop runs from two starts: SIMC [2] on a half-rule first-order-plus-dead-time model, with the closed-loop time constant three times the model dead time (cautious), and the same tuning with all gains doubled (bold). P, I and D act on the error. As in a real controller, the sampling period stays fixed, and each record covers 15 times the dead time plus the lag time constants. $M_s$ is that of this digital loop, computed from its exact $L$. Performance is measured by the load-step IAE, the integral of $|y|$ after a unit step disturbance at the plant input.

*The certificates held and were tight:* For all 276 accepted tunings, a four times longer record showed a tail below $3 \cdot 10^{-5}$, well within $\tau = 10^{-3}$, and every $M_s$ lay

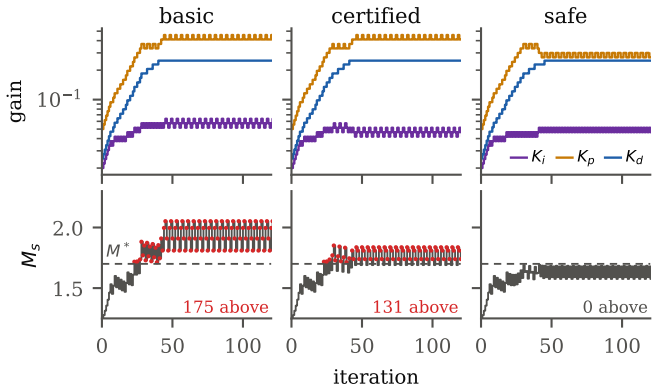



Fig. 4. Tuning history, $P_3$ with PID, cautious start, as the online simulator shows it: the gains (top) and $M_s$ of the tuning applied in each iteration (bottom; dots: $M_s > M^*$).

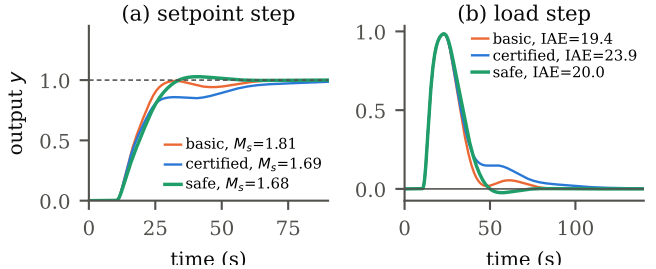



Fig. 5. The tuned loops of Fig. 4 (final tunings): output after (a) a unit setpoint step and (b) a unit load step at the plant input.

between its bounds, at most 0.003 below $\overline{M}$ (Theorem 1). Of 456 random moves from accepted tunings, with gains scaled by up to 8, all 339 certified moves were stable and all 117 rejected ones unstable, and the bound (6) exceeded $M_s$ by a median of 0.011 (Theorem 3).

*Certified SPIN ends below $M^*$, only safe SPIN stays below it:* Table III shows both corollaries. Basic SPIN ended above $M^*$ on 9 and 7 of the 19 loops; certified SPIN on none, and every loop kept accepting tunings (Corollary 1). But certified SPIN must apply a tuning to reject it, so it still applied tunings above $M^*$ in 45% of its iterations, against 57% for basic SPIN. Safe SPIN applied none, and all its tunings were stable (Corollary 2; every start tuning passed the test, so the corollary covered every iteration).

*Robustness costs little with safe SPIN:* Figs. 3–5 follow $P_3$ with PID. From the record of the basic tuning, (4) predicted the $M_s$ after a band move exactly (Fig. 3). Basic and certified SPIN climb alike (Fig. 4); certified SPIN then lowers $K_i$, but it must apply each tuning of its final cycle to reject it, so 131 of its 200 applied tunings exceed $M^*$. Safe SPIN applies only certified moves and trades $K_p$ for $K_d$. The setpoint responses are alike (Fig. 5), and the basic tuning does not overshoot although its $M_s$ is 1.81: the response alone does not reveal fragility. In load rejection, certified SPIN, with its lower $K_i$, is 23% worse than basic SPIN, and safe SPIN only 3%. Over all loops, safe and certified SPIN have the same median load-step IAE; the mean of safe SPIN is higher only because of the fast PID loop, where it settled at $M_s \approx 1.5$.

TABLE II

THE 19 TEST LOOPS

| | plant $P(s)$ | controller | loops |
|---|---|---|---|
| from [7] | $P_1 = \mathrm{e}^{-s}/((10s+1)(s+1)^3)$ | PI, PID | 2 |
| | $P_2 = 1.25\,\mathrm{e}^{-8s}/(5s+1)^4$ | PI, PID | 2 |
| | $P_3 = \mathrm{e}^{-10s}/((2s+1)(s+1)^3)$ | PI, PID | 2 |
| | $P_4 = \mathrm{e}^{-4s}/(8s+1)^6$ | PI, PID | 2 |
| one band | $\mathrm{e}^{-5s}/(s+1)$ | PI | 1 |
| | $\mathrm{e}^{-s}/(10s+1)$ | PI | 1 |
| | $\mathrm{e}^{-0.5s}/((10s+1)(s+1))$ | PI | 1 |
| | $\mathrm{e}^{-0.1s}/((s+1)(0.5s+1))$ (fast) | PID | 1 |
| family | $\mathrm{e}^{-L_d s}/((s+1)(0.5s+1)(0.25s+1))$, $L_d = 0.25, 0.5, 1, 2, 4, 8, 16$ | PID | 7 |
| total | | | 19 |

Fixed sampling period: $T_a/1000$ for PI; for PID, the smaller of $T_a/100$ and $T_f/20$ of the start tuning, where $T_a$ is the dead time plus the lag time constants and $T_f = T_d/10$, $T_d = K_d/K_p$, the derivative filter time constant.

TABLE III

RESULTS ON THE 19 LOOPS, FROM THE CAUTIOUS (C) AND BOLD (B) START

| | basic | | certified | | safe | |
|---|---|---|---|---|---|---|
| | c | b | c | b | c | b |
| *final tuning*[a] | | | | | | |
| loops with $M_s > M^*$ | 9 | 7 | 0 | 0 | 0 | 0 |
| largest $M_s$ | 2.01 | 2.07 | 1.69 | 1.69 | 1.69 | 1.69 |
| load-step IAE[b] | – | – | +9% | +8% | +15% | +22% |
| *while tuning* (3800 iterations) | | | | | | |
| iterations with $M_s > M^*$ | 2146 | 2225 | 1623 | 1775 | 0 | 0 |
| largest $M_s$ | 2.35 | 2.38 | 1.93 | 1.98 | 1.696 | 1.695 |

[a]The most aggressive tuning accepted in the last 60 iterations. [b]Mean change from basic SPIN.

*Count limits do not replace the certificate:* With every $\bar{N}_k = 1/2$, certified SPIN still applied tunings above $M^*$ in 1310 and 1420 iterations, accepted no tuning on 2 and 3 loops, and raised the median load-step IAE by 11 and 19%.

## VI. CONCLUSION

The setpoint step that a model-free tuner applies anyway certifies the robustness of the loop: its record bounds $M_s$ from both sides, without a model or a dedicated experiment, and certifies the stability and $M_s$ of the next tuning before it is applied. Checking a tuning after applying it makes the result safe; checking it before makes the whole tuning safe. The theorems assume a linear loop and a long enough record; noise and real loops are left to a follow-up. Scripts: `https://github.com/ottapav/robopid-simulator`.

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